

A PROOF OF THE AULUCK–CHOWLA–GUPTA CONJECTURE

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ABSTRACT. Let $p(n, k)$ denote the number of partitions of n into exactly k parts. In 1942 Auluck, Chowla and Gupta conjectured that for every n the sequence $p(n, 1), p(n, 2), \dots, p(n, n)$ is unimodal. Szekeres proved this for all sufficiently large n , but with a threshold that was never made explicit. We prove the conjecture for every $n \geq 1$. For $n \geq 10^5$ we give an analytic argument with fully explicit constants: an exact probabilistic representation of the ratio $p(n, k+1)/p(n, k)$, a second-order saddle-point main term with a rigorous uniform error bound in the transition window, and separate elementary or saddle-point estimates in the two tails. Every numerical constant in this argument is certified in ball arithmetic. For $n \leq 2 \cdot 10^5$ the conjecture is verified exhaustively by two independent programs, one of them using exact integer arithmetic only. Since log-concavity of $k \mapsto p(n, k)$ fails for every $n \geq 4$, the argument works with first differences and never uses log-concavity.

1. INTRODUCTION

For integers $n \geq 1$ and $1 \leq k \leq n$ let $p(n, k)$ be the number of partitions of n into exactly k positive parts. A finite sequence $a_1, \dots, a_N$ of real numbers is *(weakly) unimodal* if there is an index t with $a_1 \leq \dots \leq a_t \geq a_{t+1} \geq \dots \geq a_N$.

Main Theorem. *For every integer $n \geq 1$ the sequence $p(n, 1), p(n, 2), \dots, p(n, n)$ is unimodal.*

In fact we prove the strict form, in the shape of Szekeres’ theorem, for every n (Theorem 9.1 in Section 9): for $n \geq 4$ there is t with

$$p(n, 1) < \dots < p(n, t) \geq p(n, t+1) > p(n, t+2) > \dots > p(n, n-1) = p(n, n),$$

and $p(n, t) = p(n, t+1)$ only for $n \in \{5, 6, 8, 13, 14\}$.

History. The statement was conjectured by Auluck, Chowla and Gupta [1], who also showed that the mode $k_0(n)$ satisfies $n^{1/2} < k_0 \leq (1 + \delta)c^{-1}n^{1/2} \log n$ for large n , where $c = \pi\sqrt{2/3}$. Erdős [7] determined k_0 up to $o(n^{1/2})$ (see also Erdős–Lehner [8]) and wrote that he was unable to prove or disprove the unimodality conjecture. Szekeres [10, 11] obtained an asymptotic formula for $p(n, k)$ uniform in k , and in [11, Theorem 2] proved that *if n is sufficiently large* there is $k_1(n)$ with $p(n, k) < p(n, k+1)$ for $k < k_1$ and $p(n, k) > p(n, k+1)$ for $k > k_1$, remarking that this proves the conjecture of Auluck, Chowla and Gupta. His constants are not made explicit, so the argument gives no value of n beyond which it applies (see also [12]). Canfield [5] later gave a different derivation of Szekeres’ formula, by recursions rather than by contour integration. In a 2003 survey Canfield [4] notes that unimodality had been observed computationally for every row with $n \leq 2000$, that Szekeres’ theorem covers only sufficiently large n , and that the question is therefore “technically not completely resolved”; Balazard [3] likewise lists it as known only for large n . As recently as 2015, Chubarikov and Fedorov [6] describe the monotonicity conjecture of Auluck, Chowla and Gupta as unsolved. To our knowledge no proof for all n has appeared before.

2020 *Mathematics Subject Classification.* Primary 05A17; Secondary 11P82, 05A20, 65G30.

Key words and phrases. integer partitions, unimodality, saddle-point method, interval arithmetic, computer-assisted proof.

Why first differences. Unimodality would follow from log-concavity, $p(n, k)^2 \geq p(n, k-1)p(n, k+1)$, but this fails at the tail of every row with $n \geq 4$: for $n/2 \leq k \leq n$ one has $p(n, k) = p(n-k)$ (Lemma 7.9), so $p(n, n) = p(n, n-1) = 1$ and $p(n, n-2) = 2$, and $p(n, n-1)^2 = 1 < 2 = p(n, n)p(n, n-2)$. We therefore work directly with the sign of the first difference, through the ratio

$$R_k = R_k(n) := \frac{p(n, k+1)}{p(n, k)} \quad (1 \leq k \leq n-1).$$

The row is unimodal if and only if the signs of $\log R_k$, read left to right, never change from negative to positive.

Strategy. Put

$$(1.1) \quad \beta := \frac{\sqrt{6n}}{\pi}, \quad \ell := \log \beta, \quad \mu := \beta \ell.$$

The mode of the row lies within $O(\beta)$ of μ (this is the Gumbel scaling of Erdős and Lehner [8]). For $n \geq 10^5$ we split $1 \leq k \leq n-1$ into five ranges,

$$\begin{aligned} D &= [1, k_D], & CL &= (k_D, \mu - 2\beta], & W &= (\mu - 2\beta, \mu + 2\beta), \\ CR &= [\mu + 2\beta, \lceil n/2 \rceil], & E &= [\lceil n/2 \rceil, n-1], \end{aligned}$$

with $k_D = \lfloor 1.7n^{1/3} \rfloor$, and determine the sign of $\log R_k$ on each range separately.

- On D an elementary binomial sandwich shows $R_k > 1$ (Section 7.5).
- On E an exact bijection shows $R_k \leq 1$ (Lemma 7.9).
- Everything else rests on an exact identity (Section 2): R_k equals the ratio of two Fourier integrals of the characteristic function of a sum of independent geometric variables, taken at its saddle point θ . This reduces the problem to estimating a ratio of integrals with the *same* weight, so that all exponentially large factors cancel exactly.
- On CL and CR a zeroth-order version of this estimate is enough to give $R_k > 1$ and $R_k < 1$ respectively (Section 7).
- In the window W , where the sign changes, we need more. We define an explicit second-order main term $L_{2,k}$ (Section 4) and prove $|\log R_k - L_{2,k}| \leq 0.4282\theta q$, where $q = e^{-(k+1)\theta}$ (Theorem 4.6). We then show that L_2 decreases by at least $0.91\theta q$ at each step (Theorem 6.1). Since $0.91 > 2 \times 0.4282$, at most one $k \in W$ has an undetermined sign, and that k is adjacent to the sign change of L_2 .

The constants in these estimates are bounded by rigorous ball arithmetic (Arb [9], through python-flint) over finite covers of the relevant parameter ranges. The analytic argument is carried out for $n \geq 10^5$, and the range $n \leq 2 \cdot 10^5$ is checked exhaustively (Section 9).

Computer assistance. All computer-assisted steps are listed in Section 10 (Table 2), each with the script that certifies it and the exact statement certified. The scripts, their expected outputs and a driver that re-runs everything and compares outputs are available at <https://github.com/jaideepsaipadhi/the-auluck-chowla-gupta-conjecture>. The mathematical arguments in this paper do not depend on any computation other than those listed there.

Organisation. Section 2 gives the notation and the exact representation. Section 3 proves uniform derivative bounds. Section 4 defines the main term and states and proves the error bound of Theorem 4.6; its uniformity rests on the ingredient bounds of Section 5. Section 6 analyses the step of L_2 . Section 7 treats the tail regions, Section 8 assembles the proof for $n \geq 10^5$, Section 9 describes the finite verification, and Section 10 lists the computer-assisted inequalities and the trust base. The longer error algebra is in the appendices.

Conventions. $\zeta_2 := \zeta(2) = \pi^2/6$, so that $n = \zeta_2 \beta^2$. $\text{Li}_s(y) = \sum_{l \geq 1} y^l / l^s$ is the polylogarithm; for integers $r \geq 1$, $\text{Li}_{1-r}(y) = \sum_l l^{r-1} y^l$ is a rational function of y . Logarithms of complex numbers are principal and are only taken of numbers with positive real part. For a function g and an interval I , $\text{Var}_I g$ is the total variation of g on I , and $\text{TV}(g) = \text{Var}_{[0,\infty)} g$.

2. NOTATION AND AN EXACT REDUCTION

Fix n and $2 \leq k \leq n-2$. Write $a_k = p(n, k)$ and

$$m := n - k \quad (\geq 2), \quad K := k + 1.$$

Let $f_k(N)$ be the number of partitions of N into parts of size at most k , with $f_k(N) = 0$ for $N < 0$.

Lemma 2.1. $p(n, k) = f_k(n - k)$, and $f_{k+1}(N) = \sum_{s \geq 0} f_k(N - sK)$. Consequently $a_{k+1} = \sum_{s \geq 0} f_k(m - 1 - sK)$, a finite sum.

Proof. These are standard (see [2]). The conjugate of a partition of n into exactly k parts has largest part exactly k ; removing one part k is a bijection onto partitions of $n - k$ into parts $\leq k$. For the second identity, classify partitions of N into parts $\leq K$ by the number s of parts equal to K and remove them. Finally $a_{k+1} = f_{k+1}(m - 1)$. $\square$

2.1. The saddle point and a random variable. For $t > 0$ let

$$(2.1) \quad S_k(t) := \sum_{i=1}^k \frac{i}{e^{it} - 1}.$$

S_k is continuous and strictly decreasing from $+\infty$ to 0, so there is a unique $\theta = \theta(n, k) > 0$ with $S_k(\theta) = m$. We call θ the *saddle point* and put

$$(2.2) \quad x := e^{-\theta}, \quad q := x^K = e^{-K\theta}, \quad \nu := K\theta, \quad \nu' := \nu + \log \theta, \quad v := k\theta = \nu - \theta,$$

so that $q = \theta e^{-\nu'}$. Let $Y_1, \dots, Y_k$ be independent with $\mathbb{P}(Y_i = y) = (1 - x^i)x^{iy}$ ($y \geq 0$) and put $X := \sum_{i \leq k} iY_i$.

Lemma 2.2. $\mathbb{P}(X = N) = f_k(N)x^N \prod_{i \leq k} (1 - x^i)$, and $\mathbb{E}X = S_k(\theta) = m$.

Proof. $\mathbb{P}(X = N) = \prod_i (1 - x^i) \cdot x^N \cdot \#\{(y_i) : \sum i y_i = N\}$ and the count is $f_k(N)$, with y_i the multiplicity of the part i . Also $\mathbb{E}Y_i = x^i / (1 - x^i) = 1 / (e^{i\theta} - 1)$. $\square$

Proposition 2.3 (Exact representation). Let $\chi(\varphi) := \mathbb{E} e^{i\varphi(X-m)}$ and

$$g_0 := \frac{x}{1-q}, \quad H(\varphi) := i\varphi - \log(1 - qe^{iK\varphi}) + \log(1-q).$$

Then

$$(2.3) \quad R_k = \sum_{s \geq 0} x^{1+sK} \frac{\mathbb{P}(X = m - 1 - sK)}{\mathbb{P}(X = m)},$$

and, with $D := \int_{-\pi}^{\pi} \chi(\varphi) d\varphi = 2\pi \mathbb{P}(X = m) > 0$ and $\tilde{N} := \int_{-\pi}^{\pi} \chi(\varphi) (e^{H(\varphi)} - 1) d\varphi$,

$$(2.4) \quad \frac{R_k}{g_0} - 1 = \frac{\tilde{N}}{D}.$$

Proof. Since $f_k(m) \geq 1$ we have $\mathbb{P}(X = m) > 0$. By Lemma 2.2, $f_k(N) = \mathbb{P}(X = N)x^{-N}/\prod(1-x^i)$, and inserting this into Lemma 2.1 gives (2.3). The series $\chi(\varphi) = \sum_N \mathbb{P}(X = N)e^{i\varphi(N-m)}$ converges absolutely and uniformly, so $\frac{1}{2\pi} \int_{-\pi}^{\pi} \chi(\varphi)e^{ij\varphi} d\varphi = \mathbb{P}(X = m-j)$. Inserting this into (2.3) and exchanging the sum over s with the integral (dominated by $\sum_s x^{1+sK}$) gives $R_k = \int \chi G / \int \chi$ with $G(\varphi) = \sum_s x^{1+sK} e^{i(1+sK)\varphi} = xe^{i\varphi}/(1-qe^{iK\varphi})$. Since $\operatorname{Re}(1-qe^{iK\varphi}) \geq 1-q > 0$, H is well defined and $e^H = G/g_0$, which gives (2.4). $\square$

Remark 2.4. Identity (2.3) holds for every $x \in (0, 1)$, not only at the saddle point, because the x -dependence cancels. The saddle point is chosen so that χ is concentrated near $\varphi = 0$. The form (2.4), rather than $R_k = N/D$, matters: a relative error δ in D costs only $\delta |R_k/g_0 - 1|$, which is of order $\delta\theta^2$.

By independence,

$$(2.5) \quad \chi = e^\psi, \quad \psi(\varphi) := -im\varphi + \sum_{i \leq k} \left[\log(1-x^i) - \log(1-x^i e^{i\varphi}) \right],$$

a real-analytic function of $\varphi \in \mathbb{R}$. Since X is real-valued, $\chi(-\varphi) = \overline{\chi(\varphi)}$, so $\operatorname{Re} \psi$ is even and $\operatorname{Im} \psi$ is odd. Moreover

$$(2.6) \quad -2 \log |\chi(\varphi)| = S(\varphi) := \sum_{i \leq k} \log \left(1 + \frac{\sin^2(i\varphi/2)}{\sinh^2(i\theta/2)} \right),$$

because $|1 - ye^{ia}|^2/(1-y)^2 = 1 + \sin^2(a/2)/\sinh^2(u/2)$ for $y = e^{-u}$. Every summand of S is nonnegative.

2.2. Cumulants and the functions f_r, F_r . For $r \geq 2$ put

$$(2.7) \quad \kappa_r := \sum_{i \leq k} i^r \operatorname{Li}_{1-r}(x^i), \quad B := \kappa_2, \quad c_1 := 1 + \frac{Kq}{1-q}, \quad c_r := K^r \operatorname{Li}_{1-r}(q) \quad (r \geq 2).$$

For $r \geq 1$ let $f_r(u) := u^r \operatorname{Li}_{1-r}(e^{-u})$ for $u > 0$, extended continuously by $f_r(0) = (r-1)!$, and let $F_r(v) := \int_0^v f_r$. Thus $f_1(u) = u/(e^u - 1)$ and $f_2(u) = (u/2)^2/\sinh^2(u/2)$; both are decreasing and take values in $(0, 1]$. We use the scaled quantities

$$(2.8) \quad s_r := \theta^{r+1} \kappa_r = \theta \sum_{i \leq k} f_r(i\theta), \quad b := s_2 = \theta^3 B, \quad c_r \theta^r = f_r(\nu) \quad (r \geq 2), \quad c_1 \theta = \theta + f_1(\nu).$$

Lemma 2.5 (Closed form of F_r). *For $r \geq 1$ and $v > 0$,*

$$F_r(v) = r! \left(\zeta_2 - \sum_{j=0}^r \frac{v^j}{j!} \operatorname{Li}_{2-j}(e^{-v}) \right), \quad F_r(\infty) = r! \zeta_2.$$

In particular F_r is nondecreasing and bounded by $r! \zeta_2$.

Proof. Call the right side G_r . From $\frac{d}{dv} \operatorname{Li}_s(e^{-v}) = -\operatorname{Li}_{s-1}(e^{-v})$ and the Leibniz rule, the derivative of the sum over j telescopes to $-v^r \operatorname{Li}_{1-r}(e^{-v})/r!$, so $G'_r = f_r$. As $v \rightarrow 0^+$, $\operatorname{Li}_2(e^{-v}) \rightarrow \zeta_2$, $v \operatorname{Li}_1(e^{-v}) = -v \log(1 - e^{-v}) \rightarrow 0$, and $v^j \operatorname{Li}_{2-j}(e^{-v}) = O(v)$ for $j \geq 2$ because $\operatorname{Li}_{2-j}(e^{-v}) = O(v^{1-j})$; hence $G_r(0^+) = 0 = F_r(0)$. Since f_r is continuous on $[0, v]$, $F_r = G_r$. As $v \rightarrow \infty$ every $v^j \operatorname{Li}_{2-j}(e^{-v})$ tends to 0. $\square$

2.3. Main term. Define

$$(2.9) \quad \begin{aligned} N_0 &:= 1 + \frac{\kappa_4}{8B^2} - \frac{5\kappa_3^2}{24B^3}, & E_2 &:= \frac{1}{B} \left(1 + \frac{\kappa_4}{2B^2} - \frac{5\kappa_3^2}{4B^3} \right), \\ E_1 &:= \frac{1}{N_0} \left(\frac{\kappa_3}{2B^2} - \frac{\kappa_5}{8B^3} + \frac{35\kappa_3\kappa_4}{48B^4} - \frac{35\kappa_3^3}{48B^5} \right), \\ M_2 &:= 1 + c_1 E_1 - \frac{(c_2 + c_1^2)E_2}{2} - \frac{5c_3\kappa_3}{12B^3} + \frac{c_4}{8B^2}, \end{aligned}$$

and

$$(2.10) \quad L_k := \log R_k, \quad L_{2,k} := -\theta - \log(1-q) + \log M_2 = \log g_0 + \log M_2.$$

The origin of M_2 is explained in Section 4: it is the closed form of the ratio of the Gaussian model integrals up to the order needed. Note that $L_k - L_{2,k} = \log(R_k/(g_0 M_2))$.

3. UNIFORM DERIVATIVE BOUNDS

Lemma 3.1. $\psi(0) = \psi'(0) = 0$, $\psi^{(r)}(0) = i^r \kappa_r$ and $|\psi^{(r)}(\varphi)| \leq \kappa_r$ for all $\varphi \in \mathbb{R}$ and $r \geq 2$.

Proof. For $0 < y < 1$ and $a > 0$, $\ell_{y,a}(\varphi) := -\log(1 - ye^{ia\varphi}) = \sum_{l \geq 1} y^l e^{ila\varphi}/l$, absolutely and uniformly. The r -times differentiated series $(ia)^r \sum_l l^{r-1} y^l e^{ila\varphi}$ is dominated by $a^r \text{Li}_{1-r}(y) < \infty$, so by the Weierstrass M-test termwise differentiation is valid and $|\ell_{y,a}^{(r)}| \leq a^r \text{Li}_{1-r}(y)$. Now $\psi = -im\varphi + \sum_i [\log(1 - x^i) + \ell_{x^i,i}]$; for $r \geq 2$ the linear part disappears, which gives the bound and the value at 0. Finally $\psi'(0) = i(-m + \sum_i ix^i/(1 - x^i)) = 0$ by the saddle-point equation. $\square$

Lemma 3.2. $H(0) = 0$, $H'(0) = ic_1$, $H^{(r)}(0) = i^r c_r$ for $r \geq 2$; $|H'| \leq c_1$ and $|H^{(r)}| \leq c_r$ ($r \geq 2$) on $\mathbb{R}$; and $\text{Re } H \leq 0$.

Proof. $H = i\varphi + \ell_{q,K} + \log(1 - q)$, so $H' = i + iK \sum_l q^l e^{ilK\varphi}$ and $|H'| \leq 1 + Kq/(1 - q) = c_1$; for $r \geq 2$, $H^{(r)} = \ell_{q,K}^{(r)}$. Finally $|1 - qe^{iK\varphi}|^2 - (1 - q)^2 = 2q(1 - \cos K\varphi) \geq 0$, so $|G| \leq g_0$ and $\text{Re } H = \log |G/g_0| \leq 0$. $\square$

Corollary 3.3. Let $\omega := \psi + B\varphi^2/2$. Then $\omega(0) = \omega'(0) = \omega''(0) = 0$ and $|\omega'''| \leq \kappa_3$; hence $|\omega(\varphi)| \leq \kappa_3|\varphi|^3/6$. Moreover $|\text{Im } \psi(\varphi)| \leq \kappa_3|\varphi|^3/6$ and $\text{Re } \psi(\varphi) \leq -B\varphi^2/2 + \kappa_4\varphi^4/24$.

Proof. The first statements follow from Lemma 3.1 and Taylor's theorem with integral remainder (applied to real and imaginary parts). $\text{Im } \psi$ is odd and vanishes to second order at 0 because $\psi'(0) = 0$ and $\psi''(0) = -B$ is real. $\text{Re } \psi$ is even, with $(\text{Re } \psi)''(0) = -B$, vanishing odd derivatives at 0, and $|(\text{Re } \psi)''''| \leq \kappa_4$. $\square$

We shall also use two elementary facts repeatedly.

- (F1) If $\text{Re } z \leq 0$ then $|e^z - 1| \leq |z|$ and $|e^z - 1 - z - z^2/2| \leq |z|^3/6$; more generally $|e^z - 1| \leq |z| \max(1, e^{\text{Re } z})$. (From $e^z - 1 = z \int_0^1 e^{sz} ds$ and $e^z - 1 - z - z^2/2 = z^3 \int_0^1 \frac{(1-s)^2}{2} e^{sz} ds$.)
- (F2) For $0 \leq A$ and $0 \leq t \leq 1$, $\log(1 + At) \geq t \log(1 + A)$ (concavity).

4. THE MAIN TERM AND THE ERROR BOUND IN THE WINDOW

Throughout this section $2 \leq k \leq n - 2$ and θ is the saddle point. We fix the central arc

$$|\varphi| \leq \varphi_0 := 0.4\theta.$$

4.1. The Gaussian model. Put $g(\varphi) := e^{-B\varphi^2/2}$, and let

$$\begin{aligned}\omega_3 &= -i\kappa_3\varphi^3/6, & \omega_4 &= \kappa_4\varphi^4/24, & \omega_5 &= i\kappa_5\varphi^5/120, \\ H_1 &= i c_1\varphi, & H_2 &= -c_2\varphi^2/2, & H_3 &= -i c_3\varphi^3/6, & H_4 &= c_4\varphi^4/24,\end{aligned}$$

be the Taylor terms of $\omega = \psi + B\varphi^2/2$ and of H at 0 (Lemmas 3.1 and 3.2). Define the polynomials

$$Q := 1 + \omega_3 + \omega_4 + \omega_5 + \frac{1}{2}\omega_3^2 + \omega_3\omega_4 + \frac{1}{6}\omega_3^3, \quad P_H := 1 + H_1 + H_2 + H_3 + H_4 + \frac{1}{2}H_1^2,$$

and the model integrals

$$D_0 := \int_{\mathbb{R}} g Q \, d\varphi, \quad \tilde{N}_0 := \int_{\mathbb{R}} g Q (P_H - 1) \, d\varphi, \quad M_{\text{model}} := 1 + \tilde{N}_0/D_0.$$

They are evaluated exactly with the Gaussian moments $\int_{\mathbb{R}} \varphi^{2j} e^{-A\varphi^2/2} \, d\varphi = \sqrt{2\pi/A} (2j-1)!! A^{-j}$. In particular $D_0 = \sqrt{2\pi/B} N_0$ with N_0 as in (2.9), and $\tilde{N}_0/D_0$ is real. We work with the scaled variables

$$(4.1) \quad A_r := \frac{\kappa_r}{B^{r/2}} = \frac{s_r \theta^{r/2-1}}{b^{r/2}}, \quad C_r := \frac{c_r}{B^{r/2}} = \frac{f_r(\nu) \theta^{r/2}}{b^{r/2}} \quad (r \geq 2), \quad C_1 := \frac{c_1}{B^{1/2}},$$

and $w_0 := \varphi_0 \sqrt{B} = 0.4 \sqrt{b/\theta}$; every quantity below is invariant under $\kappa_r \mapsto \lambda^r \kappa_r$, $c_r \mapsto \lambda^r c_r$, $B \mapsto \lambda^2 B$, so it may be computed with $B = 1$ and (κ_r, c_r) replaced by (A_r, C_r) .

Lemma 4.1. $M_{\text{model}} - M_2 = \text{Num}/N_0$, where, in the scaled variables,

$$\begin{aligned}\text{Num} &= -\frac{25}{48} C_4 A_3^2 + \frac{1}{6} C_4 A_4 + (C_2 + C_1^2) \left(\frac{1}{32} A_4^2 + \frac{25}{192} A_3^4 - \frac{25}{192} A_3^2 A_4 \right) \\ &\quad + \frac{5}{4} C_3 A_3^3 + \frac{7}{48} C_3 A_5 - \frac{25}{24} C_3 A_3 A_4.\end{aligned}$$

Proof. A polynomial identity in the variables $A_3, \dots, A_5, C_1, \dots, C_4$, obtained by expanding $\tilde{N}_0$ and D_0 in Gaussian moments. It is verified symbolically (Table 2, item S2). $\square$

On the window, $f_r(\nu) = O(q) = O(\theta)$, so by (4.1) $A_r = O(\theta^{r/2-1})$, $C_r = O(\theta^{1+r/2})$ ($r \geq 2$) and $C_1 = O(\theta^{3/2})$. Hence every monomial of Num is $O(\theta^4)$, whereas $M_2 - 1 = O(\theta^2)$ is of the same order as the step θq of L_2 (Theorem 6.1); this is why a second-order main term is needed.

4.2. The pointwise error bound. Let $a_r := \kappa_r/r!$ and $t = |\varphi|$, and set

$$\delta_0 := \frac{\kappa_4 \varphi_0^2/12 + \kappa_6 \varphi_0^4/360}{B}, \quad B' := (1 - \delta_0)B.$$

Define the nonnegative polynomials in t

$$\begin{aligned}T_\omega(t) &:= a_6 t^6 + \left(\frac{1}{2} a_4^2 + a_3 a_5 \right) t^8 + a_4 a_5 t^9 + \frac{1}{2} a_5^2 t^{10} \\ &\quad + \frac{1}{2} (a_4 t^4 + a_5 t^5) (a_3 t^3 + a_4 t^4 + a_5 t^5)^2 + \frac{1}{24} (a_3 t^3 + a_4 t^4 + a_5 t^5)^4, \\ T_H(t) &:= \frac{1}{120} c_5 t^5 + \left(\frac{1}{2} c_1 c_2 + \frac{1}{6} c_1^3 \right) t^3,\end{aligned}$$

and let $Q_{\text{abs}}(t)$ and $P_{\text{abs}}(t)$ be the polynomials obtained from Q and $P_H - 1$ by replacing each coefficient by its modulus.

Lemma 4.2 (Central arc). *Suppose $\delta_0 < 1$. For $|\varphi| \leq \varphi_0$,*

$$|e^\omega - Q| \leq T_\omega(t) e^{\delta_0 B t^2/2}, \quad |e^H - P_H| \leq T_H(t), \quad |e^H - 1| \leq c_1 t,$$

and consequently

$$\begin{aligned} \left| \int_{|\varphi| \leq \varphi_0} [\chi(e^H - 1) - gQ(P_H - 1)] \right| &\leq \int_{\mathbb{R}} e^{-B't^2/2} c_1 t T_\omega(t) d\varphi + \int_{\mathbb{R}} g Q_{\text{abs}} T_H d\varphi, \\ \left| \int_{|\varphi| \leq \varphi_0} (\chi - gQ) \right| &\leq \int_{\mathbb{R}} e^{-B't^2/2} T_\omega(t) d\varphi. \end{aligned}$$

Proof. By Taylor's theorem with $|\psi^{(6)}| \leq \kappa_6$ and since the odd-order terms are purely imaginary, $\text{Re } \omega \leq \kappa_4 t^4/24 + \kappa_6 t^6/720 \leq \delta_0 B t^2/2$ for $t \leq \varphi_0$. Let $\hat{\omega} = \omega_3 + \omega_4 + \omega_5$; then $|\omega - \hat{\omega}| \leq a_6 t^6$ and $\text{Re } \hat{\omega} = \kappa_4 t^4/24 \leq \delta_0 B t^2/2$. We split

$$e^\omega - Q = (e^\omega - e^{\hat{\omega}}) + \left(e^{\hat{\omega}} - \sum_{j \leq 3} \frac{\hat{\omega}^j}{j!} \right) + \left(\sum_{j \leq 3} \frac{\hat{\omega}^j}{j!} - Q \right).$$

The first term is at most $|\omega - \hat{\omega}| e^{\max(\text{Re } \omega, \text{Re } \hat{\omega})}$, the second is at most $\frac{|\hat{\omega}|^4}{24} \max(1, e^{\text{Re } \hat{\omega}})$ (Taylor's theorem with integral remainder), and the third is the polynomial $\frac{1}{2}(\omega_4^2 + \omega_5^2) + \omega_3 \omega_5 + \omega_4 \omega_5 + \frac{1}{6}(\hat{\omega}^3 - \omega_3^3)$, whose modulus is dominated termwise by the remaining terms of T_ω . For T_H , write $e^H - P_H = (e^H - 1 - H - \frac{1}{2}H^2) + (H - \sum_{r \leq 4} H_r) + \frac{1}{2}(H - H_1)(H + H_1)$. By Lemma 3.2, $|H| \leq c_1 t$, so (F1) bounds the first piece by $c_1^3 t^3/6$; Taylor's theorem with $|H^{(5)}| \leq c_5$ bounds the second by $c_5 t^5/120$; and $|H - H_1| \leq c_2 t^2/2$, $|H + H_1| \leq 2c_1 t$ bound the third by $c_1 c_2 t^3/2$. The bound $|e^H - 1| \leq c_1 t$ is (F1). The integral inequalities follow from the identity $\chi(e^H - 1) - gQ(P_H - 1) = g[(e^\omega - Q)(e^H - 1) + Q(e^H - P_H)]$ and from $g e^{\delta_0 B t^2/2} = e^{-B't^2/2}$. $\square$

The *model tails* are bounded by $e^{-B\varphi^2/2} \leq e^{-B\varphi_0^2/4} e^{-B\varphi^2/4}$ for $|\varphi| \geq \varphi_0$:

$$(4.2) \quad \begin{aligned} \int_{|\varphi| > \varphi_0} g |Q(P_H - 1)| &\leq e^{-B\varphi_0^2/4} \int_{\mathbb{R}} e^{-B\varphi^2/4} Q_{\text{abs}} P_{\text{abs}}, \\ \int_{|\varphi| > \varphi_0} g |Q| &\leq e^{-B\varphi_0^2/4} \int_{\mathbb{R}} e^{-B\varphi^2/4} Q_{\text{abs}}. \end{aligned}$$

On the *minor arcs* $\varphi_0 \leq |\varphi| \leq \pi$, since $|e^H - 1| \leq 2$ by $\text{Re } H \leq 0$ and the arcs have total length at most 2π ,

$$(4.3) \quad \left| \int_{\varphi_0 \leq |\varphi| \leq \pi} \chi(e^H - 1) \right| \leq 4\pi M_\chi, \quad \left| \int_{\varphi_0 \leq |\varphi| \leq \pi} \chi \right| \leq 2\pi M_\chi,$$

where $M_\chi := \sup_{\varphi_0 \leq |\varphi| \leq \pi} |\chi(\varphi)|$.

Proposition 4.3 (Pointwise error bound). *Let E_N (resp. E_D) be the sum of the numerator (resp. denominator) bounds of Lemma 4.2, (4.2) and (4.3), all divided by $\sqrt{2\pi/B}$ (recall $D_0/\sqrt{2\pi/B} = N_0$). Suppose $\delta_0 < 1$, $N_0 - E_D > 0$ and*

$$\Delta := \frac{E_N + |\tilde{N}_0/D_0| E_D}{N_0 - E_D} + \frac{|\text{Num}|}{N_0} < M_2.$$

Then

$$(4.4) \quad |L_k - L_{2,k}| \leq \varepsilon_k := -\log(1 - \Delta/M_2).$$

Proof. Write $\tilde{N} = \tilde{N}_0 + e_N$ and $D = D_0 + e_D$. The three groups of estimates give $|e_N| \leq \sqrt{2\pi/B} E_N$ and $|e_D| \leq \sqrt{2\pi/B} E_D$. Since $\tilde{N}/D - \tilde{N}_0/D_0 = (e_N - (\tilde{N}_0/D_0)e_D)/(D_0 + e_D)$, Proposition 2.3 gives $|R_k/g_0 - M_{\text{model}}| \leq \Delta - |\text{Num}|/N_0$, and Lemma 4.1 gives $|R_k/(g_0 M_2) - 1| \leq \Delta/M_2 < 1$. Finally $|\log(1+y)| \leq -\log(1-|y|)$ for $|y| < 1$, and $L_k - L_{2,k} = \log(R_k/(g_0 M_2))$. $\square$

All the integrals in E_N and E_D are integrals of nonnegative polynomials against Gaussians over $\mathbb{R}$, which we evaluate exactly via $\int_{\mathbb{R}} |\varphi|^p e^{-A\varphi^2/2} d\varphi = (2/A)^{(p+1)/2} \Gamma(\frac{p+1}{2})$. The full expression for ε_k , written in the scaled variables (4.1), is recorded in Appendix A. It is a fixed algebraic expression in $(\theta, \nu', b, s_3, \dots, s_6)$ and in an upper bound for M_χ ; all of its algebra (the moments, the polynomial identities above, and the scaling invariance) is checked symbolically (Table 2, S2).

4.3. The minor arcs. The only non-local input is a bound for M_χ . The following lemma gives it uniformly in θ ; its proof uses only the partial sums of $\sin^2$, and no counting arguments.

Lemma 4.4 (Minor arcs). *Let $\theta_{\max} > 0$, $u_{\text{top}} > 0$ and $c_0 > 0$, and suppose $\theta \leq \theta_{\max}$ and $(k+1)\theta \geq u_{\text{top}}$. Let $c_a \geq c_0$ with $c_a \theta_{\max}/2 \leq \pi/2$, and put $L_* := 2\theta_{\max} + 2/(c_a \text{sinc}(c_a \theta_{\max}/2))$.*

(a) *If $\varphi/\theta \in [c_{\text{lo}}, c_{\text{hi}}] \subset [c_0, c_a]$, then for every $T \in (0, \pi/2]$, with $\sigma = \sin T/T$,*

$$\theta S(\varphi) \geq \int_{\theta_{\max}}^{\min(u_{\text{top}}, 2T/c_{\text{hi}})} \log(1 + \sigma^2 c_{\text{lo}}^2 f_2(u)) du.$$

(b) *If $c_a \theta \leq \varphi \leq \pi$, then $\theta S(\varphi) \geq \int_{L_*}^{u_{\text{top}}} \log \coth(u/2) du$ (to be read as 0 if $L_* \geq u_{\text{top}}$).*

Consequently $\log |\chi(\varphi)| \leq -\Psi/(2\theta)$ for $c_0 \theta \leq |\varphi| \leq \pi$, where Ψ is the minimum of the right-hand sides over a cover of $[c_0, c_a]$ by cells and of (b).

Proof. By (2.6) and $\chi(-\varphi) = \overline{\chi(\varphi)}$ we may take $\varphi > 0$; recall that every summand of S is ≥ 0 .

(a) Put $c = \varphi/\theta$. For $i \leq 2T/\varphi$ we have $i\varphi/2 \leq T$, so $\sin(i\varphi/2) \geq \sigma i\varphi/2$, because $\sin y/y$ is decreasing on $(0, \pi)$. Hence the i -th summand is at least $\log(1 + \sigma^2 c^2 (i\theta/2)^2 / \sinh^2(i\theta/2)) \geq \lambda(i\theta)$ with $\lambda(u) := \log(1 + \sigma^2 c_{\text{lo}}^2 f_2(u))$, a decreasing function. Keep only $i \leq I := \min(k, \lfloor 2T/\varphi \rfloor)$. Then $\theta S \geq \theta \sum_{i \leq I} \lambda(i\theta) \geq \int_{\theta}^{(I+1)\theta} \lambda$. Since $(\lfloor 2T/\varphi \rfloor + 1)\theta > 2T/c$, we have $(I+1)\theta \geq \min((k+1)\theta, 2T/c) \geq \min(u_{\text{top}}, 2T/c_{\text{hi}})$, and $\theta \leq \theta_{\max}$, $\lambda \geq 0$.

(b) Put $d = \varphi/2 \in (0, \pi/2]$ and $w_i := 2 \log \coth(i\theta/2) = \log(1 + 1/\sinh^2(i\theta/2))$, which is decreasing in i . By (F2) with $t = \sin^2(id)$, the i -th summand is at least $\sin^2(id) w_i$. Let $G_I := \sum_{i \leq I} \sin^2(id)$. The Dirichlet identity

$$(4.5) \quad G_I = \frac{I}{2} - \frac{\sin(Id) \cos((I+1)d)}{2 \sin d}$$

gives $G_I \geq I/2 - 1/(2 \sin d)$, and since also $G_I \geq 0$ we get $G_I \geq \frac{1}{2}(I - N_d)_+$ with $N_d := \lceil 1/\sin d \rceil$. By Abel summation, with $w_{k+1} := 0$ and $w_I - w_{I+1} \geq 0$,

$$S \geq \sum_{i \leq k} \sin^2(id) w_i = \sum_{I \leq k} G_I (w_I - w_{I+1}) \geq \frac{1}{2} \sum_{I \leq k} (I - N_d)_+ (w_I - w_{I+1}) = \frac{1}{2} \sum_{N_d < i \leq k} w_i.$$

As w is decreasing, $w_i \geq \theta^{-1} \int_{i\theta}^{(i+1)\theta} 2 \log \coth(u/2) du$, so $\theta S \geq \int_{(N_d+1)\theta}^{(k+1)\theta} \log \coth(u/2) du$. Finally $(N_d+1)\theta \leq 2\theta + \theta/\sin d$, and since $\sin$ is increasing on $[0, \pi/2]$ and sinc is decreasing, $\sin d \geq \sin(c_a \theta/2) \geq (c_a \theta/2) \text{sinc}(c_a \theta_{\max}/2)$. Hence $(N_d+1)\theta \leq L_*$ and $(k+1)\theta \geq u_{\text{top}}$. $\square$

Corollary 4.5. *If $\theta \leq 0.0042$ and $(k+1)\theta \geq 3$, then $|\chi(\varphi)| \leq e^{-c_*/\theta}$ for $0.4\theta \leq |\varphi| \leq \pi$, with $c_* = 0.15973$.*

Proof. Apply Lemma 4.4 with $\theta_{\max} = 0.0042$, $u_{\text{top}} = 3$, $c_0 = 0.4$ and $c_a = 0.4 \cdot 150^{96/400} = 1.33144\dots$; then $L_* \leq 1.5105$. All integrals are right Riemann sums of decreasing integrands, hence rigorous lower bounds, evaluated in ball arithmetic: (a) gives $\theta S \geq 0.31946$ on the 96 geometric cells of ratio $150^{1/400}$ covering $[0.4, c_a]$ (best of 41 values of T per cell), and (b) gives $\theta S \geq 0.34433$. So $\Psi \geq 0.31946$ and $c_* = \Psi/2 \geq 0.15973$ (Table 2, C3). $\square$

4.4. Statement of Theorem A.

Theorem 4.6 (Theorem A). *Let $n \geq 10^5$ and $k \in W = (\mu - 2\beta, \mu + 2\beta)$. Then*

$$|L_k - L_{2,k}| \leq \varepsilon_k \leq \Gamma_{\max} \theta q, \quad \Gamma_{\max} = 0.4282.$$

More precisely $\varepsilon_k \leq 0.42024 \theta q$ when $\theta \geq 0.002$, and $\varepsilon_k \leq 0.17605 \theta q$ when $\theta \leq 0.002$.

The first inequality is Proposition 4.3. The second is proved in Section 5.3 after the ingredient bounds are in place.

5. INGREDIENT BOUNDS

5.1. The window. Write $n = \zeta_2 \beta^2$ and $\beta_A := \sqrt{6 \cdot 10^5} / \pi = 246.56 \dots$, $\ell_A = \log \beta_A$.

Lemma 5.1 (Lemma W). *Let $n \geq 10^5$, $k \in W$, and put $\delta := 5\ell/\beta$, $\epsilon_\beta := (\ell + 2 + 1/\beta)/(\zeta_2 \beta)$ and $r := (1 - \epsilon_\beta)^{-1/2}$. Then*

$$\frac{1 - \delta}{\beta} < \theta \leq \frac{r}{\beta},$$

and consequently

$$\theta \leq \theta_A := 0.0040939, \quad \nu' \in [-2.5102, 2.0840], \quad \nu \geq 3.1158.$$

Proof. Two-sided bound for $t^2 S_k(t)$. We have $t^2 S_k(t) = t \sum_{i \leq k} f_1(it)$ with f_1 decreasing and $0 < f_1 \leq 1$. Comparing with integrals, $F_1((k+1)t) - F_1(t) \leq t^2 S_k(t) \leq F_1(kt) \leq \zeta_2$, and $F_1(t) \leq t$. Write $F_1(v) = \zeta_2 - T(v)$ with $T(v) = \int_v^\infty f_1 = \sum_l e^{-lv}(v/l + 1/l^2) \leq (v+1)e^{-v}/(1 - e^{-v})$.

Upper bound. Since $k < \mu + 2\beta$, $m > n - \beta(\ell + 2)$, and $\theta^2 m \leq \zeta_2$ gives $\theta^2 \leq \zeta_2/m \leq r^2/\beta^2$.

Lower bound. Let $\theta_- := (1 - \delta)/\beta$. Since S_k is decreasing it suffices to show $S_k(\theta_-) > m$. Now $\theta_-^2 m \leq \theta_-^2 n = (1 - \delta)^2 \zeta_2$ and $\theta_-^2 S_k(\theta_-) \geq \zeta_2 - T(K\theta_-) - \theta_-$, so it suffices that $(2\delta - \delta^2)\zeta_2 > T(K\theta_-) + \theta_-$. As $K > \beta(\ell - 2)$, $K\theta_- \geq (\ell - 2)(1 - \delta) =: v_-$, and the bound for T turns the sufficient condition into $G > 0$, where (after multiplication by β/ℓ)

$$G := 2 \cdot 5 \zeta_2 - 25 \zeta_2 \ell / \beta - e^2 \beta^\delta / (1 - e^{-v_-}) - 1/\ell.$$

Parameter ranges. At fixed K , $\nu' = K\theta + \log \theta$ is increasing in θ . With $K < \mu + 2\beta + 1$ and $\theta \leq r/\beta$ this gives $\nu' \leq (\ell + 2 + 1/\beta)r - \ell + \log r$; with $K > \mu - 2\beta$ and $\theta > (1 - \delta)/\beta$ it gives $\nu' \geq -2 - \delta(\ell - 2) + \log(1 - \delta)$ and $\nu \geq (\ell - 2)(1 - \delta)$.

Certification. Each of the resulting expressions, and G , is a function of $x = 1/\beta$, ℓx , $\ell^2 x$ and $1/\ell$. An adaptive ball-arithmetic bisection over $\ell \in [\ell_A, 60]$ (5486 boxes), together with one tail box for $\ell \geq 60$ (where $x, \ell x, \ell^2 x \leq 3600e^{-60}$ since $\ell^a e^{-\ell}$ is decreasing for $\ell > a$), certifies $G \geq 1.0092$ and the stated constants for every $\beta \geq \beta_A$ (Table 2, C1). No monotonicity in β is used. $\square$

The same certification gives the constants used for the other regions:

$$(5.1) \quad \nu' \leq \nu_{CL} := -1.95361 \text{ for } k \leq \mu - 2\beta, \quad \nu' \geq \nu_{CR} := 1.04305 \text{ for } k \geq \mu + 2\beta,$$

and the edge values $\nu' \leq -1.906674$ at the left end of W and $\nu' \geq 1.04305$ at the right end. (For (5.1) one uses that θ_k is increasing in k , since $S_k(t)$ increases with k while $m = n - k$ decreases.)

5.2. Riemann sums.

Lemma 5.2. *Let $f : [0, \infty) \rightarrow \mathbb{R}$ have $\text{TV}(f) < \infty$. Then for $\theta > 0$ and $k \geq 1$, $|\theta \sum_{i \leq k} f(i\theta) - \int_0^{k\theta} f| \leq \theta \text{Var}_{[0, k\theta]} f \leq \theta \text{TV}(f)$.*

Proof. $\theta f(i\theta) - \int_{(i-1)\theta}^{i\theta} f = \int_{(i-1)\theta}^{i\theta} (f(i\theta) - f(u)) du$, and $|f(i\theta) - f(u)| \leq \text{Var}_{[(i-1)\theta, i\theta]} f$. Summing over i and using additivity of the variation gives the claim. $\square$

Lemma 5.3. *The total variations of $f_1, \dots, f_6$ on $[0, \infty)$ satisfy*

$$\text{TV}(f_1), \dots, \text{TV}(f_6) \leq 1.000001, 1.000001, 2.000001, 6.086765, 25.691943, 138.425.$$

Proof. On $[60, \infty)$ each f_r is decreasing to 0 (each term $l^{r-1}u^r e^{-lu}$ decreases for $u > r$), so the variation there is $f_r(60)$. On $[0, 60]$ we use 6000 cells. On a cell where ball arithmetic certifies that f'_r has constant sign, f_r is monotone and the variation is $|f_r(b) - f_r(a)|$. Otherwise the cell is bisected (to depth 14), and on the final cells the variation is bounded by $(b-a) \sup |f'_r| \geq \int_a^b |f'_r|$. For $u \leq 2.5$, f_r and f'_r are evaluated from their Bernoulli-number expansions with a rigorous tail bound ($|B_j|/j! \leq 4/(2\pi)^j$); for $u > 2.5$, from the polylogarithm and the identity $u f'_r = r f_r - f_{r+1}$ (Table 2, C2). $\square$

Corollary 5.4. $s_r \in F_r(\nu - \theta) \pm \theta \text{TV}(f_r)$ and $0 \leq s_r \leq r! \zeta_2 + \theta \text{TV}(f_r)$. In particular $b \geq F_2(\nu - \theta) - \theta \text{TV}(f_2)$.

Proof. By (2.8), $s_r = \theta \sum_{i \leq k} f_r(i\theta)$; apply Lemmas 5.2 and 2.5 with $k\theta = \nu - \theta$. $\square$

Note that $f_r(\nu) = c_r \theta^r$ and $c_1 = 1 + \nu e^{-\nu'} / (1 - q)$ are *exact* functions of (θ, ν') . Thus every quantity in Proposition 4.3 is enclosed, given (θ, ν') , by the enclosures of Corollary 5.4, and M_χ by Corollary 4.5.

5.3. Proof of Theorem A. By Lemma 5.1, for $k \in W$ we have $\theta \leq \theta_A$, $\nu' \in [-2.52, 2.10]$ and $\nu \geq 3.1158$; in particular Corollary 4.5 applies. Since $\theta q = \theta^2 e^{-\nu'}$, it suffices to bound $\Gamma := \varepsilon_k / (\theta^2 e^{-\nu'})$.

Region I: $0.002 \leq \theta \leq \theta_A$. We cover θ by geometric cells of ratio 1.01 and ν' by cells of width 0.01 (clipped to $\nu \geq 3.1158$). On each box we evaluate ε of Proposition 4.3 in ball arithmetic, with s_r ranging over the enclosure of Corollary 5.4 for every (θ, ν') in the box, $f_r(\nu)$ and c_1 exact, and $M_\chi \leq e^{-0.15926/\theta}$ (a rounded-down value, valid since $0.15926 \leq c_*$); the code asserts $\delta_0 < 1$, $N_0 - E_D > 0$ and $\Delta < M_2$ on every box, and divides by $\min \theta^2 e^{-\nu'}$ over the box. The maximum is $\Gamma \leq 0.42024$, attained near $\nu' = 2.1$ (Table 2, A1).

Region II: $\theta \leq 0.002$. Here no box cover is possible, and we use a monotone majorant in $t = \sqrt{\theta}$. After dividing by $\theta q = t^2 q$, every term of ε is bounded by a nonnegative combination of monomials $t^P \nu^j e^{-\beta q \nu'}$ with coefficients that are bounded uniformly through $s_r \leq r! \zeta_2 + 0.002 \text{TV}(f_r)$, $b \geq F_2(\nu_{\min} - 0.002) - 0.002 \text{TV}(f_2)$ and $e^{-\nu} \leq e^{-\nu_{\min}}$. At fixed ν' we have $\nu = \nu' + 2 \log(1/t)$ and

$$\frac{d}{dt} [t^P \nu^j] = t^{P-1} \nu^{j-1} (P\nu - 2j),$$

which is nonnegative when $P > 0$ and $\nu \geq 2j/P$ (and trivially when $j < 0$). The program asserts, for every monomial, either $P = j = 0$ (a constant term) or $P > 0$ and $\nu_{\min} > 2j/P$, and evaluates the majorant at $\theta = 0.002$. The Gaussian-tail and minor-arc factors $e^{-0.04b/\theta} \theta^{-2}$ and $e^{-c_*/\theta} \theta^{-3.5}$ are increasing on the relevant range because $\theta < 0.02b$ and $\theta < c_*/3.5$, which is asserted. The result is $\Gamma \leq 0.17605$ for all $\theta \leq 0.002$ and $\nu' \in [-2.52, 2.10]$ (Table 2, A2). The details of the majorant are in Appendix B. $\square$

6. THE STEP OF THE MAIN TERM

In this section k and $k+1$ both lie in W , and primes refer to the index $k+1$: $\theta' = \theta_{k+1}$ is the saddle point for $(m-1, k+1)$, $q' = e^{-(K+1)\theta'}$, $\nu_{k+1} = (K+1)\theta'$, and so on.

Theorem 6.1 (Theorem B). *Let $n \geq 10^5$ and $k, k+1 \in W$. Then*

$$L_{2,k} - L_{2,k+1} \geq \Sigma_{\min} \theta_k q_k, \quad \Sigma_{\min} = 0.91918,$$

and $\theta_{k+1} q_{k+1} \leq 1.01 \theta_k q_k$. In particular L_2 is strictly decreasing on W and

$$L_{2,k} - L_{2,k+1} \geq 0.9100 \max(\theta_k q_k, \theta_{k+1} q_{k+1}) > 2\Gamma_{\max} \max(\theta_k q_k, \theta_{k+1} q_{k+1}).$$

6.1. Decomposition.

$$(6.1) \quad L_{2,k} - L_{2,k+1} = (\theta' - \theta) + \log \frac{1 - q'}{1 - q} - (\log M'_2 - \log M_2).$$

We shall see that $\theta' > \theta$, so the first term is positive and $q' = e^{-(K+1)\theta'} < qe^{-\theta}$. Hence

$$\log \frac{1 - q'}{1 - q} \geq T_2 := \log \left(1 + \frac{q(1 - e^{-\theta})}{1 - q} \right),$$

which is $\theta q(1 + O(\theta + q))$, and the step is at least $T_2 - |\log M'_2 - \log M_2|$. The work is to bound $\log M'_2 - \log M_2$, for which we need to control how $(\theta, \nu, b, s_3, s_4, s_5)$ moves from k to $k+1$.

6.2. Localising the next saddle point. Let $B_k(t) := -S'_k(t) = t^{-3}b(k, t)$ with $b(k, t) := t \sum_{i \leq k} f_2(it)$. Note $S_{k+1}(t) - S_k(t) = K/(e^{Kt} - 1)$, which equals $c_1 - 1$ at $t = \theta$. Fix $\tau = 0.01$ and let $b_{\min} := F_2(\nu - \theta) - (1 + \tau)\theta \text{TV}(f_2)$. By Corollary 5.4 and monotonicity of F_2 , $b(k, t) \geq b_{\min}$ for $t \in [\theta, (1 + \tau)\theta]$.

Lemma 6.2. (i) *If $\tau b_{\min} > (1 + \tau)^3 c_1 \theta^2$, then $\theta < \theta' \leq (1 + \tau)\theta$.*

(ii) *Under (i), $d\theta := \theta' - \theta \leq (1 + \tau)^3 c_1 \theta^3 / b_{\min}$.*

(iii) *$d\nu := \nu_{k+1} - \nu = \theta' + K d\theta \in (0, (1 + \tau)\theta + (\nu/\theta) d\theta]$.*

(iv) *For $r = 2, \dots, 5$, with $\bar{s}_r := r! \zeta_2 + (1 + \tau)\theta \text{TV}(f_r)$,*

$$|s_r(k+1, \theta') - s_r(k, \theta)| \leq d\theta \frac{(1+r)\bar{s}_r + \bar{s}_{r+1}}{\theta} + (1 + \tau)\theta \sup_{u \in [\nu, \nu + d\nu]} f_r(u).$$

Proof. (i) $S_{k+1}(\theta) = m + c_1 - 1 > m - 1 = S_{k+1}(\theta')$, so $\theta' > \theta$. On $[\theta, (1 + \tau)\theta]$, $-S'_{k+1} \geq B_k(t) \geq b_{\min}/((1 + \tau)\theta)^3$, so $S_{k+1}((1 + \tau)\theta) \leq m + c_1 - 1 - \tau b_{\min}/((1 + \tau)^3 \theta^2) < m - 1$. (ii) By the mean value theorem $c_1 = S_{k+1}(\theta) - S_{k+1}(\theta') = B_{k+1}(\xi) d\theta \geq b_{\min} d\theta/((1 + \tau)\theta)^3$. (iii) $\nu_{k+1} - \nu = (K + 1)\theta' - K\theta$ and $K = \nu/\theta$. (iv) $s_r(k+1, \theta') = s_r(k, \theta') + \theta' f_r(K\theta')$ with $K\theta' \in [\nu, \nu + d\nu]$. From $u f'_r = r f_r - f_{r+1}$ we get $\partial_t s_r(k, t) = ((1 + r)s_r(k, t) - s_{r+1}(k, t))/t$, and $0 \leq s_r(k, t) \leq \bar{s}_r$ for $t \leq (1 + \tau)\theta$. $\square$

6.3. Proof of Theorem B. Region I ($0.002 \leq \theta \leq \theta_A$). We cover (θ, ν') by 8526 boxes (θ -ratio 1.02, ν' -width 0.02, clipped to $\nu \geq 3.1158$). On each box the program asserts the hypothesis of Lemma 6.2(i) and encloses $d\theta$, $d\nu$ and the increments of s_r . It then forms a product box $\mathcal{H} \subset \mathbb{R}^6$ in the variables $(\theta, \nu, b, s_3, s_4, s_5)$: $\theta \in [\theta, (1 + \tau)\theta]$, $\nu \in [\nu, \nu + d\nu]$ and $s_r \in [F_r(\nu - \theta) - (1 + \tau)\theta \text{TV}_r, F_r(\nu + d\nu) + (1 + \tau)\theta \text{TV}_r]$. Both the point for k and the point for $k+1$ lie in $\mathcal{H}$ (Corollary 5.4 and Lemma 6.2), and $\mathcal{H}$ is convex. The program checks $M_2 > 0$ on $\mathcal{H}$ and encloses ∇M_2 on $\mathcal{H}$ by forward-mode automatic differentiation in ball arithmetic, so by the mean value theorem along the segment

$$|\log M'_2 - \log M_2| \leq \sum_j \sup_{\mathcal{H}} \left| \frac{\partial_j M_2}{M_2} \right| |v'_j - v_j|,$$

where $v = (\theta, \nu, b, s_3, s_4, s_5)$ and v' is the same vector at $k+1$, with $|v'_j - v_j|$ bounded by Lemma 6.2. Together with $T_2/(\theta q)$ evaluated on the box this gives $\Sigma := (\text{step})/(\theta q) \geq 0.91918$ on every box (Table 2, B1).

Region II ($\theta \leq 0.002$). The same graded-monomial method as in Section 5.3 applies. At both points every quantity is majorised by nonnegative combinations of $t^p q^{b_q} \nu^j$, using $\theta'^e \leq (1+\tau)^e \theta^e$ ($e > 0$), $q' \leq q$, $\nu_{k+1} \leq \rho \nu$ with an explicit ρ , $f_r(u) \leq (\rho \nu)^r q P_{r-1}(y)/(1-y)^r$ ($y = e^{-\nu_{\min}}$, P_{r-1} the Eulerian polynomial), and $|h'(u)| \leq (u+1)e^{-u}/(1-e^{-u})^2$ for $c_1 = 1 + h(\nu)/\theta$, $h(u) = u/(e^u - 1)$. Products and quotients are handled by $|x'y' - xy| \leq |x' - x| \sup |y| + \sup |x| |y' - y|$ and $|1/N' - 1/N| \leq |N' - N|/N_{\min}^2$ (primes denoting values at $k+1$), and the main term by $T_2/(\theta q) \geq (1 - \theta/2)(1 - \theta q/(2(1-q)))$. The result is $\Sigma \geq 0.96039$ for all $\theta \leq 0.002$ and $\nu' \in [-2.52, 2.10]$, and also $|\log M_2|/\theta \leq 0.14215$ (Table 2, B2).

Finally $\theta' q' \leq 1.01 \theta q$ because $\theta' \leq 1.01 \theta$ and $q' < q$, and $0.91918/1.01 > 0.9100 > 0.8564 = 2\Gamma_{\max}$. $\square$

6.4. Consequences in the window. Let k^* be the largest $k \in W$ with $L_{2,k} > 0$, and $k^* := \min(W \cap \mathbb{Z}) - 1$ if there is no such k (so $L_{2,k} \leq 0$ for every $k \in W$ with $k > k^*$). Call $k \in W$ *undetermined* if $|L_{2,k}| \leq \varepsilon_k$. If k is not undetermined then $\text{sign } L_k = \text{sign } L_{2,k}$ by Theorem 4.6.

Corollary 6.3. *Let $n \geq 10^5$. At most one $k \in W$ is undetermined, and if there is one it belongs to $\{k^*, k^* + 1\}$. Hence the signs of L_k for $k \in W$, read left to right, have the form $+\dots + - \dots -$ (either block may be empty).*

Proof. For $k \leq k^* - 1$, Theorem 6.1 gives $L_{2,k} \geq L_{2,k} - L_{2,k+1} > 0.91 \theta_k q_k > \Gamma_{\max} \theta_k q_k \geq \varepsilon_k$. For $k \geq k^* + 2$, $L_{2,k} \leq L_{2,k} - L_{2,k-1} \leq -0.91 \theta_k q_k$. So only k^* and $k^* + 1$ can be undetermined. If both were, then

$$L_{2,k^*} - L_{2,k^*+1} \leq \varepsilon_{k^*} + \varepsilon_{k^*+1} \leq 2\Gamma_{\max} \max(\theta q) < 0.91 \max(\theta q),$$

contradicting Theorem 6.1. The determined signs are $+$ for $k \leq k^*$ and $-$ for $k \geq k^* + 1$, and a single undetermined index adjacent to the change cannot create a second change. $\square$

Remark 6.4 (Location of the mode). Using the edge values of ν' after Lemma 5.1 one finds $L_2/\theta \geq 5.588$ at the left end of W and $L_2/\theta \leq -0.505$ at the right end (Table 2, B3), so the sign change of L_2 , and hence the mode of the row, lies inside W . This is not needed for unimodality.

7. THE REGIONS D, CL, CR AND E

7.1. A zeroth-order estimate. Outside the window a much cruder version of Proposition 4.3 suffices. Let $c_0 > 0$ and $\varphi_0 = c_0 \theta$.

Lemma 7.1 (Lemma C0). *Suppose $\text{Re } \psi(\varphi) \leq -(1-\delta)B\varphi^2/2$ for $|\varphi| \leq \varphi_0$, with $0 \leq \delta < 1$. Put*

$$T_1 = \frac{c_1 \kappa_3}{2B^2}, \quad T_2 = \frac{c_2 + c_1^2}{2B}, \quad A_3 = \kappa_3 B^{-3/2}, \quad w_0 = \varphi_0 \sqrt{B}, \quad \mathcal{M} = \sqrt{\frac{B}{2\pi}} \int_{\varphi_0 \leq |\varphi| \leq \pi} |\chi|.$$

If the denominator below is positive, then

$$\left| \frac{R_k}{g_0} - 1 \right| \leq \rho := \frac{T_1(1-\delta)^{-5/2} + T_2(1-\delta)^{-3/2} + 2\mathcal{M}}{1 - \text{erfc}(w_0/\sqrt{2}) - \sqrt{2/\pi} A_3/(3(1-\delta)^2) - \mathcal{M}}.$$

Moreover $\mathcal{M} \leq \sqrt{2\pi B} \sup_{\varphi_0 \leq |\varphi| \leq \pi} |\chi|$.

Proof. We use (2.4) and measure all integrals in units of $\sqrt{2\pi/B}$. *Numerator, central arc.* Since $(e^H)'' = (H'' + H'^2)e^H$ and $\operatorname{Re} H \leq 0$, Taylor's theorem gives $e^H - 1 = ic_1\varphi + E(\varphi)$ with $|E| \leq (c_2 + c_1^2)\varphi^2/2$. By $\chi(-\varphi) = \overline{\chi(\varphi)}$, $\int_{|\varphi| \leq \varphi_0} \chi\varphi = 2i \int_0^{\varphi_0} \varphi \operatorname{Im} \chi$, and $|\operatorname{Im} \chi| = e^{\operatorname{Re} \psi} |\sin \operatorname{Im} \psi| \leq e^{-(1-\delta)B\varphi^2/2} \kappa_3 |\varphi|^3/6$ by Corollary 3.3. Integrating φ^4 and φ^2 against $e^{-(1-\delta)B\varphi^2/2}$ over $\mathbb{R}$ gives the terms $T_1(1-\delta)^{-5/2}$ and $T_2(1-\delta)^{-3/2}$. *Numerator, minor arcs.* $|e^H - 1| \leq 2$ gives $2\mathcal{M}$. *Denominator.* D is real and positive. On $|\varphi| \leq \varphi_0$ write $\chi = e^{-B\varphi^2/2}e^\omega$; by Corollary 3.3 and (F1), $|e^\omega - 1| \leq |\omega|e^{\delta B\varphi^2/2} \leq \frac{\kappa_3|\varphi|^3}{6}e^{\delta B\varphi^2/2}$, whose weighted integral is $\sqrt{2/\pi}A_3/(3(1-\delta)^2)$. Also $\int_{|\varphi| \leq \varphi_0} e^{-B\varphi^2/2} = \sqrt{2\pi/B}(1 - \operatorname{erfc}(w_0/\sqrt{2}))$, and the minor arcs contribute at most $\mathcal{M}$. $\square$

Lemma 7.2 (Admissible δ). *Both $\delta_T := \kappa_4\varphi_0^2/(12B)$ and, when $c_0k\theta < 2$, $\delta_C := 1 - (1 - (c_0k\theta)^2/12) \log(1 + c_0^2)/c_0^2$ satisfy the hypothesis of Lemma 7.1 (when they are < 1).*

Proof. For δ_T use Corollary 3.3: $\operatorname{Re} \psi \leq -B\varphi^2/2 + \kappa_4\varphi_0^2\varphi^2/24$. For δ_C , by (2.6), $-2\operatorname{Re} \psi = \sum_i \log(1 + \sin^2(i\theta/2)/\sinh^2(i\theta/2))$. For $i \leq k$ and $|\varphi| \leq c_0\theta$ the angle $y = i|\varphi|/2$ is < 1 , and $\sin^2 y \geq (y - y^3/6)^2 \geq y^2(1 - y^2/3) \geq \lambda y^2$ with $\lambda := 1 - (c_0k\theta)^2/12$. With $a_i := (i\varphi/2)^2/\sinh^2(i\theta/2) \leq c_0^2$, two applications of (F2) give $\log(1 + \lambda a_i) \geq \lambda \log(1 + a_i) \geq \lambda a_i \log(1 + c_0^2)/c_0^2$, and $\sum_i a_i = B\varphi^2$. $\square$

From $\log R_k = -\theta - \log(1 - q) + \log(R_k/g_0)$, $q \leq -\log(1 - q) \leq q/(1 - q)$ and $1 - q \leq \nu$ we obtain, whenever $\rho < 1$,

$$(7.1) \quad \log R_k \leq -\theta + \frac{q}{1 - q} + \rho, \quad \log R_k \geq -\theta + q + \log(1 - \rho), \quad \log R_k \geq -\theta - \log \nu + \log(1 - \rho).$$

7.2. A bound uniform in θ .

Lemma 7.3 (Lemma C1). *Fix $0 < v_1 < v_2 \leq \infty$, $c_0 > 0$ and $\bar{\theta} > 0$. Suppose $\Psi > 5\bar{\theta}$ is such that $\theta S(\varphi) \geq \Psi$ for $c_0\theta \leq |\varphi| \leq \pi$ whenever $\theta \leq \bar{\theta}$ and $k\theta \in [v_1, v_2]$. Then there is an explicit number $X = X(\bar{\theta}, v_1, v_2, c_0)$ such that $\rho \leq \theta X$ for every saddle point $\theta \leq \bar{\theta}$ with $k\theta \in [v_1, v_2]$.*

Proof. Write $v = k\theta$. We bound the ingredients of Lemma 7.1:

- $s_r \leq \min(F_r(v_2) + \theta \operatorname{TV}(f_r), v \sup f_r)$ for $r = 3, 4$ (Corollary 5.4, and each term $\theta f_r(i\theta) \leq \theta \sup f_r$); with $\sup f_3 \leq 2.0302009$, $\sup f_4 \leq 6.1253939$;
- $b \geq \max(F_2(v_1) - \theta, v_1 f_2(v_2))$ and $b \leq F_2(v_2)$, by comparing the decreasing $f_2 \leq 1$ with its integral;
- $c_1\theta = \theta + f_1(\nu) \leq \theta + f_1(v_1)$ and $c_2\theta^2 = f_2(\nu) \leq f_2(v_1)$, as $\nu \geq v \geq v_1$;
- $\sup |\chi| \leq e^{-\Psi/(2\theta)}$, hence $\mathcal{M} \leq \sqrt{2\pi b} \theta^{-3/2} e^{-\Psi/(2\theta)}$.

In scaled form $T_1 = \theta(c_1\theta)s_3/(2b^2)$, $T_2 = \theta(c_2\theta^2 + (c_1\theta)^2)/(2b)$, $A_3 = s_3\theta^{1/2}b^{-3/2}$, $w_0 = c_0(b/\theta)^{1/2}$ and $\delta_T = c_0^2s_4/(12b)$. The quantity ρ is increasing in $T_1, T_2, \mathcal{M}, \delta, A_3$ and $\operatorname{erfc}(w_0/\sqrt{2})$ (its numerator increases and its denominator decreases in each). After substituting the bounds above, T_1/θ , T_2/θ , A_3 and δ are nondecreasing in θ and w_0 is nonincreasing; this is checked by computing each derivative symbolically as a polynomial with nonnegative coefficients over a positive denominator (Table 2, M). Finally $\frac{d}{d\theta} \log(\mathcal{M}/\theta) = (\Psi - 5\theta)/(2\theta^2) > 0$ for $\theta < \Psi/5$. Hence ρ/θ is bounded by its value with all ingredients taken at $\bar{\theta}$, which defines X . (The program takes the best of $c_0 \in \{0.4, 0.6, 0.8, 1, 1.25, 1.5, 2\}$ and of δ_T, δ_C .) $\square$

In this section Ψ is provided by Lemma 4.4 with $u_{\text{top}} = v_1$ (since $(k+1)\theta \geq k\theta \geq v_1$).

7.3. The right tail CR.

Proposition 7.4. *Let $n \geq 10^5$ and $\mu + 2\beta \leq k \leq \lceil n/2 \rceil$. Then $\log R_k \leq -0.614\theta < 0$.*

Proof. Since $m \geq n - \lceil n/2 \rceil \geq (n-1)/2$ and $\theta^2 m \leq \zeta_2$, $\theta \leq \theta_{CR} := \sqrt{2\zeta_2/(n-1)} \leq 0.0057358$. By (5.1), $\nu' \geq \nu_{CR} = 1.04305$, so $v = \nu' - \log \theta - \theta \geq 1.04305 + \log(1/\theta_{CR}) - \theta_{CR} \geq 6.198$; both bounds improve with n . Lemma 4.4 with $\theta_{\max} = \theta_{CR}$ and $u_{\text{top}} = 6.198$ gives $\Psi = 0.33808 > 5\theta_{CR}$ (`ws_mech_cd`), and Lemma 7.3 on the single cell $[6.198, \infty)$ (where only $F_r(\infty) = r!\zeta_2$ is used) gives $X \leq 0.0321$. By (7.1), with $q/\theta = e^{-\nu'}$ and $q \leq e^{-v}$,

$$\log R_k \leq \theta \left(\frac{e^{-\nu'}}{1-q} + X - 1 \right) \leq \theta(0.3852 - 1).$$

(Table 2, R1.) □

7.4. The left tail CL.

Proposition 7.5 (CL-a). *For every n , if $47 \leq k \leq n-2$ and $k\theta \leq 0.1$, then $R_k > 1$.*

Proof. Put $K_0 = 47$, $V_0 = 0.1$, $\theta_0 = V_0/K_0$, so $\theta \leq \theta_0$. Here $b \geq v f_2(V_0)$, $s_r \leq v \sup f_r$, $c_1 \theta \leq 1 + \theta_0$ and $f_2 \leq 1$, which give $T_1 \leq (1 + \theta_0) \sup f_3 / (2K_0 f_2(V_0)^2)$, $T_2 \leq (1 + (1 + \theta_0)^2) / (2K_0 f_2(V_0))$, $A_3 \leq \sup f_3 K_0^{-1/2} f_2(V_0)^{-3/2}$ and $w_0 \geq c_0 (K_0 f_2(V_0))^{1/2}$. We bound $\mathcal{M} \leq \mathcal{N}_1 + \mathcal{N}_2 + \mathcal{N}_{34}$ according to $\varphi \in [c_0\theta, 10\theta]$, $[10\theta, \pi/k]$ and $[\pi/k, \pi]$; since $10\theta = 10v/k < \pi/k$ these cover the minor arcs.

$\mathcal{N}_1$. Split $c = \varphi/\theta \in [c_0, 10]$ into 60 cells $[c_j, c_{j+1}]$. As $i\varphi/2 \leq c_{j+1}V_0/2 =: T_j < \pi/2$, $\sin(i\varphi/2) \geq \sigma_j i\varphi/2$ with $\sigma_j = \sin T_j/T_j$, and by (F2) each summand of S is at least $\lambda_j c^2 f_2(i\theta)$ with $\lambda_j = \log(1 + \sigma_j^2 c_{j+1}^2)/c_{j+1}^2$. Since $\sum_i c^2 f_2(i\theta) = B\varphi^2$, this gives $|\chi| \leq e^{-\lambda_j B\varphi^2/2}$, and $2\sqrt{B/2\pi} \int_{c_j\theta}^\infty e^{-\lambda_j B\varphi^2/2} d\varphi = \text{erfc}(c_j\theta\sqrt{\lambda_j B/2})/\sqrt{\lambda_j}$ with $\theta^2 B = b/\theta \geq K_0 f_2(V_0)$.

$\mathcal{N}_2$. Here every angle $i\varphi/2$ is at most $\pi/2$, hence $\sin(i\varphi/2) \geq i\varphi/\pi$, and with $c = \varphi/\theta$ each summand of S is at least $\log(4c^2 f_2(V_0)/\pi^2)$. Hence, using $\theta\sqrt{B} \leq \sqrt{k}$,

$$|\chi| \leq \left(\frac{4f_2(V_0)}{\pi^2} \right)^{-k/2} c^{-k}, \quad \mathcal{N}_2 \leq \sqrt{\frac{2}{\pi}} \sqrt{k} \left(\frac{4f_2(V_0)}{\pi^2} \right)^{-k/2} \frac{10^{1-k}}{k-1}.$$

$\mathcal{N}_{34}$. Put $d = \varphi/2 \in [\pi/(2k), \pi/2]$. By (F2) each summand is at least $\sin^2(id)w_i$ with $w_i = \log(1 + 1/\sinh^2(i\theta/2)) \geq w_k$, and by (4.5) $\sum_{i \leq k} \sin^2(id) \geq k/2 - 1/(2 \sin d)$. Since $\sin d \geq (\pi/2k) \text{sinc}(\pi/(2K_0))$, this is at least $2gk$ with $g := \frac{1}{2}(\frac{1}{2} - 1/(\pi \text{sinc}(\pi/(2K_0)))) > 0$. Also $w_k \geq -2 \log \sinh(v/2) \geq -2 \log(v\sigma_0/2)$ with $\sigma_0 = \sinh(V_0/2)/(V_0/2)$. So $|\chi| \leq (v\sigma_0/2)^{2gk}$, and with $B \leq k^3/v^2$, $\mathcal{N}_{34} \leq \sqrt{2\pi} k^{3/2} v^{-1} (v\sigma_0/2)^{2gk}$, which is increasing in v and decreasing in k (both asserted), so it is evaluated at (K_0, V_0) .

With $c_0 = 0.6$ and $\delta = \min(\delta_T, \delta_C)$ this gives $\mathcal{N}_1 \leq 3.7 \cdot 10^{-4}$, $\mathcal{N}_2 \leq 10^{-37}$, $\mathcal{N}_{34} \leq 6.4 \cdot 10^{-8}$ and $\rho \leq 0.0673$. By the third inequality of (7.1), $\log R_k > 0$ as soon as $1 - \rho > e^\theta \nu$, and $e^\theta \nu \leq e^{\theta_0} V_0 (1 + 1/K_0) \leq 0.1024$ (Table 2, R2). □

Proposition 7.6 (CL-b). *Let $n \geq 10^5$ and $k_D < k \leq \mu - 2\beta$ with $k\theta \geq 0.1$. Then $R_k > 1$.*

Proof. Here $k \geq 79$, $\theta \leq \theta_A$ and $\nu' \leq \nu_{CL} = -1.95361$ by (5.1). Cover $v = k\theta \in [0.1, \infty)$ by the 32 geometric cells $[v_1, v_2]$ of ratio 1.15 starting at 0.1 and the final cell $[0.1 \cdot 1.15^{32}, \infty) = [8.7565 \dots, \infty)$. On each cell $\theta \leq \bar{\theta} := \min(\theta_A, v_2/78)$; let X be the constant of Lemma 7.3. By (7.1), $\log R_k \geq -\theta + q - \theta X/(1 - \theta X)$, which is positive as soon as $q/\theta > 1 + X/(1 - \bar{\theta} X)$. For q/θ we use the larger of $e^{-\nu'} \geq e^{1.95361}$ and $e^{-(v_2 + \bar{\theta})/\bar{\theta}}$. Every cell passes, with minimum ratio 5.82 (Table 2, R3). □

7.5. Small k : the region D.

Lemma 7.7. For $1 \leq k \leq n$, $\binom{n-1}{k-1} \leq k! p(n, k) \leq \binom{n-1+k(k-1)/2}{k-1}$.

Proof. Sorting a composition of n into k positive parts gives a partition into k parts, and each partition arises from at most $k!$ compositions. Conversely $\lambda \mapsto (\lambda_i + k - i)_{i \leq k}$ maps partitions of n into k parts injectively to partitions of $n + k(k-1)/2$ into k distinct parts, each of which arises from exactly $k!$ compositions. $\square$

Proposition 7.8. Let $n \geq 10^5$ and $1 \leq k \leq k_D(n) := \lfloor 1.7n^{1/3} \rfloor$. Then $p(n, k+1) > p(n, k)$.

Proof. Let $\Phi(k, n) := \log(n-k) - \log(k(k+1)) - k(k-1)^2/(2(n-k+1))$. *Step 1:* $\Phi > 0$ implies the claim. By Lemma 7.7 it suffices that $\binom{n-1}{k}/\binom{N-1}{k-1} > k+1$ with $N = n + D$, $D = k(k-1)/2$. The left side equals $\frac{n-k}{k} \prod_{j=0}^{k-2} (1 + D/(n-1-j))^{-1} \geq \frac{n-k}{k} \exp(-(k-1)D/(n-k+1))$, by $1+y \leq e^y$ and $n-1-j \geq n-k+1$.

Step 2: monotonicity in k . For real $1 \leq k < n$,

$$\partial_k \Phi = -\frac{1}{n-k} - \frac{1}{k} - \frac{1}{k+1} - \frac{(k-1)[(3k-1)(n-k+1) + k(k-1)]}{2(n-k+1)^2} < 0.$$

Put $\gamma = 1.7$ and $s = n^{1/3} \geq s_A := 10^{5/3}$; since $\gamma s < n-1$, $\Phi(k, n) \geq \Phi(\gamma s, s^3)$ for $k \leq \gamma s$.

Step 3: reduction to one variable. Using $k(k-1)^2/(n-k+1) \leq k^3/(n-k)$, which follows from $k^3(n-k+1) - k(k-1)^2(n-k) = k[(n-k)(2k-1) + k^2] \geq 0$, we get $\Phi(\gamma s, s^3) \geq \Psi_D(s)$, where

$$\Psi_D(s) := \log \frac{s^2 - \gamma}{\gamma(\gamma s + 1)} - \frac{\gamma^3}{2(1 - \gamma/s^2)},$$

and, exactly,

$$\Psi'_D(s) = \frac{\gamma s^2 + 2s + \gamma^2}{(s^2 - \gamma)(\gamma s + 1)} + \frac{\gamma^4 s}{(s^2 - \gamma)^2} > 0 \quad (s > \sqrt{\gamma}).$$

Step 4. $\Psi_D(s_A) = 0.30456 \dots > 0$ (ball arithmetic), so $\Psi_D(s) > 0$ for all $s \geq s_A$ (Table 2, R4). $\square$

7.6. Large k : the region E.

Lemma 7.9. If $2k \geq n$ then $p(n, k) = p(n-k)$. Consequently $p(n, k+1) \leq p(n, k)$ for $\lceil n/2 \rceil \leq k \leq n-1$, with equality only for $k = n-1$.

Proof. Subtracting 1 from each part is a bijection from partitions of n into exactly k parts to partitions of $n-k$ into at most k parts, and the latter condition is vacuous when $n-k \leq k$. For $k \geq \lceil n/2 \rceil$ also $2(k+1) \geq n$, so $p(n, k+1) = p(j)$ and $p(n, k) = p(j+1)$ with $j = n-k-1$. Appending a part 1 injects partitions of j into those of $j+1$, and misses $(j+1)$ when $j \geq 1$. $\square$

8. PROOF OF THE MAIN THEOREM FOR $n \geq 10^5$

Let $n \geq 10^5$. The five ranges

$D = [1, k_D]$, $CL = (k_D, \mu-2\beta]$, $W = (\mu-2\beta, \mu+2\beta)$, $CR = [\mu+2\beta, \lceil n/2 \rceil]$, $E = [\lceil n/2 \rceil, n-1]$ cover $[1, n-1]$, because $\mu+2\beta = \beta(\ell+2) \leq n/2-1$: indeed $\beta(\ell+2)/(n/2) = (2/\zeta_2)(\ell+2)e^{-\ell}$ is decreasing for $\ell > -1$ and is below 0.04 at $n = 10^5$. (If $k_D \geq \mu-2\beta$, the range CL is empty and nothing changes.) Every k in $CL \cup W \cup CR$ satisfies $79 \leq k \leq \lceil n/2 \rceil \leq n-2$, so Proposition 2.3 applies there. The hypotheses of every lemma used hold for every $n \geq 10^5$; this bookkeeping is collected in Table 1 and checked by a separate program (Table 2, T).

Now:

- $R_k > 1$ for $k \in D$ (Proposition 7.8);
- $R_k > 1$ for $k \in CL$: such k satisfy $k \geq 79 \geq 47$, so Proposition 7.5 applies if $k\theta \leq 0.1$ and Proposition 7.6 if $k\theta \geq 0.1$;
- on W the signs of $\log R_k$ have the form $+\dots+ - \dots -$ (Corollary 6.3);
- $R_k < 1$ for $k \in CR$ (Proposition 7.4) and $R_k \leq 1$ for $k \in E$ (Lemma 7.9).

Concatenating, the signs of $p(n, k+1) - p(n, k)$, $1 \leq k \leq n-1$, change at most once, from nonnegative to nonpositive. Hence the row is unimodal. $\square$

TABLE 1. Hypotheses of the lemmas and why they hold for all $n \geq 10^5$.

Lemma	Hypothesis	Holds because
Cor. 4.5 (minor arcs, window)	$\theta \leq 0.0042$, $(k+1)\theta \geq 3$	Lemma 5.1: $\theta \leq 0.0040939$, $\nu \geq 3.1158$
Thm. 4.6, Region I	$\theta \in [0.002, \theta_A]$, $\nu' \in [-2.52, 2.10]$, $\nu \geq 3.1158$	Lemma 5.1
Thm. 4.6, Region II	$\theta \leq 0.002$, $\nu' \in [-2.52, 2.10]$, $\theta < c_*/3.5$	Lemma 5.1; $0.15926/3.5 > 0.002$
Thm. 6.1	same ranges at k ; hypothesis of Lemma 6.2(i)	asserted on every box and cell
Cor. 6.3	$\Sigma_{\min}/1.01 > 2\Gamma_{\max}$	$0.91008 > 0.8564$
Prop. 7.4	$\theta \leq \theta_{CR}$, $v \geq 6.198$, $\Psi > 5\theta_{CR}$	θ_{CR} decreasing in n ; $0.33808 > 0.0287$
Prop. 7.5	$k \geq 47$, $k\theta \leq 0.1$	independent of n
Prop. 7.6	$k \geq 79$, $\theta \leq \theta_A$, $\nu' \leq \nu_{CL}$	$k_D(n) \geq 78$; (5.1)
Prop. 7.8	$n \geq 10^5$, $k \leq 1.7n^{1/3}$	Ψ_D increasing
Prop. 2.3	$2 \leq k \leq n-2$	$79 \leq k \leq \lceil n/2 \rceil$ on $CL \cup W \cup CR$

9. FINITE VERIFICATION FOR $n \leq 2 \cdot 10^5$

By Lemma 7.9, the row n is unimodal if and only if the signs of $p(n, k) - p(n, k-1)$ for $2 \leq k \leq \lceil n/2 \rceil + 1$ contain no $+$ after a $-$. All $p(n, k)$ with $n \leq N$ are produced simultaneously by the column recurrence

$$v[j] \leftarrow v[j] + v[j-k] \quad (j = k, k+1, \dots), \quad k = 1, 2, \dots,$$

started from $v = (1, 0, 0, \dots)$: after step k , $v[j]$ is the number of partitions of j into parts $\leq k$, which equals $p(j+k, k)$ by Lemma 2.1. The cost is $O(N^2)$ additions.

Two independent programs implement this.

- (1) *Exact integers* (`ws_fc_exact.c`). Fixed-width multi-limb unsigned integers (27 limbs of 64 bits for $N = 2 \cdot 10^5$, justified by $p(N) < e^{\pi\sqrt{2N/3}}$; a carry beyond the last limb aborts the run). Only additions and comparisons are performed. Since $v[j+1]$ is not yet updated when $v[j]$ has just been updated at step k , the comparison of $p(n, k)$ with $p(n, k-1)$ needs no second array. The program uses no floating point and no threads.
- (2) *Interval arithmetic* (`ws_fc_check.c`). Two copies of the array in x87 extended precision (64-bit significand, no overflow since $\log_2 p(2 \cdot 10^5) \approx 1660$), one updated with rounding toward $-\infty$ and one toward $+\infty$. Since every operation is an addition of nonnegative numbers, induction gives $\text{lo}[j] \leq v[j] \leq \text{hi}[j]$ throughout. A pair is classified as $+$ or $-$ only if the intervals are disjoint; no pair remained unclassified.

For $N = 2 \cdot 10^5$ both programs report no violation; there are 256 705 214 increasing pairs, 9 743 394 777 decreasing pairs and 8 ties (all for small n). They write a per- n log (mode and verdict) and the two logs are byte-identical. For $N = 2000$ both logs also agree byte-for-byte with a third, Python big-integer computation via $p(n, k) = p(n-1, k-1) + p(n-k, k)$. As an illustration, the mode of the row is $k = 2042$ for $n = 2 \cdot 10^5$.

Since $2 \cdot 10^5 > 10^5$, Sections 8 and 9 together prove the Main Theorem for every $n \geq 1$.

Theorem 9.1 (Strict form). *For every $n \geq 4$ there is an index $t = t(n)$ with $p(n, 1) < \dots < p(n, t)$, $p(n, t) \geq p(n, t+1)$, $p(n, t+1) > p(n, t+2) > \dots > p(n, n-1)$ and $p(n, n-1) = p(n, n) = 1$. Moreover $p(n, t) = p(n, t+1)$ exactly for $n \in \{5, 6, 8, 13, 14\}$.*

Proof. For $n \leq 2 \cdot 10^5$ both programs classify every pair $(p(n, k), p(n, k+1))$ with $k \leq \lceil n/2 \rceil$ as increasing, decreasing or tied. They report exactly 8 ties, and a direct computation for $n \leq 200$ locates all eight:

$$(n, k) = (2, 1), (3, 1), (3, 2), (5, 2), (6, 2), (8, 3), (13, 4), (14, 4),$$

each between the two largest terms of its row. Lemma 7.9 gives strict decrease for $\lceil n/2 \rceil \leq k \leq n-2$. For $n \geq 10^5$, every comparison is strict in Regions D , CL and CR (Section 7) and on W outside the at most one undetermined index of Corollary 6.3; that index lies at the sign change, so a tie can only occur between the two largest terms. Region E is handled by Lemma 7.9. $\square$

10. THE COMPUTER-ASSISTED PARTS

10.1. Method. All inequalities in the analytic part that involve explicit numbers are certified in Arb [9] ball arithmetic (python-flint 0.9.0). A ball enclosure of a function on a box contains every value the function takes on that box, so a strict inequality certified on each box of a cover holds on the union. Where a statement is about all θ near 0 or all $\beta \geq \beta_A$, it is reduced to finitely many boxes either by a tail box in variables that are monotone on the tail (Lemma 5.1), or by a monotone majorant whose monotonicity is checked symbolically (Section 5.3, Lemma 7.3). Floating-point grids are used only to choose the cells. Integrals are bounded below by right Riemann sums of decreasing integrands. Exact algebraic identities (Gaussian moments, the polynomial identity of Lemma 4.1, derivatives, the Dirichlet identity (4.5), the telescoping in Lemma 2.5) are checked with SymPy.

10.2. List of certified statements. Table 2 lists every computer-assisted statement used in the proof, the program that certifies it (in the directory `analytic/` or `finite/` of the repository), and the certified value. The driver script `run_all.sh` runs all the analytic programs and the exact finite check (to $2 \cdot 10^5$ with the option `--full`) and compares each output with a stored reference; the interval program F2 is run separately (see the repository README); on a two-core machine the analytic part takes about five minutes and the exact finite check to $2 \cdot 10^5$ about eight minutes.

Table 2: Computer-assisted statements. “Sanity” rows test statements that are proved in the text and are not part of the proof.

Item	Program	Statement certified	Value
S1	<code>ws_mech_elem</code>	elementary monotonicity facts; Dirichlet identity (4.5); Abel summation; $uf'_r = rf_r - f_{r+1}$	exact (SymPy)

Item	Program	Statement certified	Value
S2	ws_mech_lemA	Gaussian moments; $D_0 = \sqrt{2\pi/\bar{B}}N_0$; $\tilde{N}_0$ real; Lemma 4.1; scaling invariance; the decompositions in Lemma 4.2; algebraic reduction to G in Lemma 5.1; independent re-implementation of ε agrees with the production code	exact; agreement $1.7 \cdot 10^{-13}$
S3	ws_close_elem	telescoping in Lemma 2.5 and its limits; Arb quadrature of F_r ; identities in Lemma 6.2; Gaussian integrals in Lemmas 7.1, 7.2, Prop. 7.5	exact / Arb
C1	ws_mech_window	Lemma 5.1 and (5.1) for all $\beta \geq \beta_A$	$G \geq 1.0092$, $\theta_A \leq 0.0040939$, $\nu' \in [-2.5102, 2.0840]$, $\nu \geq 3.1158$, $\nu_{CL} \leq -1.95361$, $\nu_{CR} \geq 1.04305$
C2	ws_mech_tv	Lemma 5.3	see Lemma 5.3
C3	ws_mech_minor	Corollary 4.5	$c_* \geq 0.15973$
A1	ws_p12_boxes	Theorem 4.6, $\theta \in [0.002, \theta_A]$	$\Gamma \leq 0.42024$
A2	ws_p12_asym	Theorem 4.6, $\theta \leq 0.002$	$\Gamma \leq 0.17605$
B1	ws_b_step	Theorem 6.1, $\theta \in [0.002, \theta_A]$	$\Sigma \geq 0.91918$
B2	ws_b_asym	Theorem 6.1, $\theta \leq 0.002$	$\Sigma \geq 0.96039$
B3	ws_b_edges, ws_mech_edges	edge values of L_2/θ (not needed for unimodality)	≥ 5.588 ; ≤ -0.505
M	ws_mech_mono	monotonicity in θ of the majorants in Lemma 7.3 and Section 5.3	exact (SymPy)
R1	ws_mech_cd, ws_cd_CR	Proposition 7.4	$0.3852 < 1$
R2	ws_mech_cla, ws_cd_CLa	Proposition 7.5	$\rho \leq 0.0673 < 0.8976$
R3	ws_mech_cd, ws_cd_CLb	Proposition 7.6	min. ratio $5.82 > 1$
R4	ws_close_D, ws_cd_D	Proposition 7.8: $\partial_k \Phi$, Ψ'_D in closed form, $\Psi_D(s_A) > 0$	$\Psi_D(s_A) = 0.30456$
T	ws_close_thresh	Table 1 and the cover in Section 8, reading the values certified above	—
F1	ws_fc_exact.c	Section 9, exact integers, $n \leq 2 \cdot 10^5$	0 violations
F2	ws_fc_check.c	Section 9, directed-rounding intervals, $n \leq 2 \cdot 10^5$	0 violations, identical log
Sanity	ws_close_rep	(2.3) in exact rational arithmetic (4959 cases), (2.4) by quadrature, Lemmas 3.1, 3.2, 7.7, 7.9 on instances	—
Sanity	ws_b_check, ws_p12_window, ws_cd_window, ws_p12_minor, ws_b_minorM	pointwise cross-checks of the above with independent code	—

10.3. Trust base. The proof relies on: the correctness of Arb’s ball arithmetic as exposed by python-flint 0.9.0 (including its polylogarithm, zeta, gamma, error-function and integration

routines); SymPy 1.14 for exact symbolic identities; the GCC 13.3 compiler for the integer code of the finite check (cross-checked by the independent interval program and, for $n \leq 2000$, by Python big integers); and the following classical results, which are quoted rather than proved: Taylor's theorem with integral remainder, the mean value theorem, the Weierstrass M-test for termwise differentiation, and additivity of total variation. All other steps are proved in this paper.

APPENDIX A. THE ERROR BOUND IN SCALED FORM

We record the expression for ε_k of Proposition 4.3 as it is evaluated. Set $B = 1$ and replace κ_r, c_r by A_r, C_r of (4.1) and φ_0 by $w_0 = 0.4\sqrt{b/\theta}$. Write $a_r = A_r/r!$, $\delta_0 = A_4 w_0^2/12 + A_6 w_0^4/360$ and $B' = 1 - \delta_0$. For a polynomial $P(t) = \sum p_j t^j$ with $p_j \geq 0$ and $A > 0$ put

$$\mathcal{I}_A[P] := \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} P(|w|) e^{-Aw^2/2} dw = \sum_j p_j \frac{2^{j/2} \Gamma(\frac{j+1}{2})}{\sqrt{\pi}} A^{-(j+1)/2}.$$

Let $Q_{\text{abs}} = 1 + a_3 t^3 + a_4 t^4 + a_5 t^5 + \frac{1}{2} a_3^2 t^6 + a_3 a_4 t^7 + \frac{1}{6} a_3^3 t^9$ and $P_{\text{abs}} = C_1 t + \frac{1}{2} (C_2 + C_1^2) t^2 + \frac{1}{6} C_3 t^3 + \frac{1}{24} C_4 t^4$. Then

$$E_N = \mathcal{I}_{B'}[C_1 t T_\omega] + \mathcal{I}_1[Q_{\text{abs}} T_H] + e^{-w_0^2/4} \mathcal{I}_{1/2}[Q_{\text{abs}} P_{\text{abs}}] + 2\mathfrak{m},$$

$$E_D = \mathcal{I}_{B'}[T_\omega] + e^{-w_0^2/4} \mathcal{I}_{1/2}[Q_{\text{abs}}] + \mathfrak{m},$$

where T_ω, T_H are as in Section 4 with $(\kappa_r, c_r) \rightarrow (A_r, C_r)$, and

$$\mathfrak{m} := 2\pi e^{-c_*/\theta} \left(\frac{b}{2\pi\theta^3} \right)^{1/2}$$

is the minor-arc term $2\pi M_\chi / \sqrt{2\pi/B}$ with Corollary 4.5. (Since $\mathcal{I}_A$ integrates against $e^{-Aw^2/2}$, the central-arc terms carry the factor $B'^{-1/2}$ and the tail terms the factor $\sqrt{2}$ automatically; the code evaluates $\mathcal{I}_A[P] = A^{-1/2} \mathbb{E}P(|Z_A|)$ with $Z_A \sim N(0, 1/A)$.) With N_0, M_2 and Num in scaled form,

$$\Delta = \frac{E_N + |\tilde{N}_0/D_0| E_D}{N_0 - E_D} + \frac{|\text{Num}|}{N_0}, \quad \varepsilon_k = -\log(1 - \Delta/M_2).$$

With the orders of magnitude noted after Lemma 4.1, $E_N = O(\theta^{7/2})$, $E_D = O(\theta^2)$, $\tilde{N}_0/D_0 = O(\theta^2)$ and $\text{Num} = O(\theta^4)$ (the tails and minor arcs being exponentially small), so $\varepsilon_k = O(\theta^{7/2})$, which is small compared with $\theta q \asymp \theta^2$ for ν' in a bounded range. Theorem 4.6 quantifies this.

APPENDIX B. THE MONOTONE MAJORANT FOR $\theta \leq 0.002$

Let $\theta_1 = 0.002$, $t = \sqrt{\theta} \leq t_1 = \sqrt{\theta_1}$, and fix a cell $[v_0, v'_0]$ of ν' . By (4.1) and $q = \theta e^{-\nu'} = t^2 e^{-\nu'}$:

- $A_r = s_r b^{-r/2} t^{r-2}$, where $s_r \leq r! \zeta_2 + \theta_1 \text{TV}(f_r)$ and $b \geq F_2(\nu_{\min} - \theta_1) - \theta_1 \text{TV}(f_2)$;
- $C_r = f_r(\nu) b^{-r/2} t^r$ with $f_r(\nu) = \nu^r e^{-\nu} P_{r-1}(e^{-\nu}) / (1 - e^{-\nu})^r \leq \nu^r q P_{r-1}(y) / (1 - y)^r$, $y = e^{-\nu_{\min}}$ (Eulerian polynomials P_{r-1} have nonnegative coefficients);
- $C_1 = c_1 (\theta^3/b)^{1/2}$ with $c_1 = 1 + \nu e^{-\nu'} / (1 - q)$.

Each term of E_N, E_D, Num and $M_2 - 1$ is a polynomial with nonnegative coefficients in these quantities, except for the explicitly signed terms of M_2 and N_0 , which are bounded below termwise (for instance $N_0 \geq 1 - \frac{5}{24} A_3^2$). Expanding, each term is a nonnegative coefficient times $t^p q^{b_q} \nu^j$. After division by $\theta q = t^2 q$ the monomial becomes $t^P \nu^j e^{-(b_q-1)\nu'}$, and at fixed $\nu', \nu = \nu' + 2 \log(1/t)$. By the identity in Section 5.3, $t \mapsto t^P \nu^j$ is nondecreasing on $(0, t_1]$ when

$P > 0$ and $\nu_{\min} > 2j/P$ (for $j < 0$ it is increasing whenever $P > 0$). The program asserts these two conditions for every non-constant monomial, evaluates each at $t = t_1$ with ν at the appropriate end ($\nu \leq \nu'_0 + \log(1/\theta_1)$ if $j \geq 0$, $\nu \geq \nu_{\min}$ if $j < 0$), and bounds $e^{-(b_q-1)\nu'}$ at the appropriate end of the cell. The non-polynomial factors $e^{-w_0^2/4} = e^{-0.04b/\theta}$ and $e^{-c_*/\theta}$ are handled by $\frac{d}{d\theta} \log(\theta^{-p}e^{-a/\theta}) = (a - p\theta)/\theta^2$, which is positive for $\theta < a/p$. The same method, applied to the increments of Lemma 6.2 at the two points k and $k + 1$, gives Region II of Theorem 6.1.

Acknowledgments. AI tools (Anthropic’s Claude) were used in this work, including for the computations, the accompanying code, and the preparation of the manuscript. The author takes full responsibility for the content, and all computational claims can be reproduced from the public repository.

REFERENCES

- [1] F. C. Auluck, S. Chowla and H. Gupta, On the maximum value of the number of partitions of n into k parts, *J. Indian Math. Soc. (N.S.)* **6** (1942), 105–112.
- [2] G. E. Andrews, *The Theory of Partitions*, Encyclopedia of Mathematics and its Applications, Vol. 2, Addison-Wesley, Reading, MA, 1976.
- [3] M. Balazard, Quelques exemples de suites unimodales en théorie des nombres, *J. Théor. Nombres Bordeaux* **2** (1990), 13–30.
- [4] E. R. Canfield, Integer partitions and the Sperner property, *Theoret. Comput. Sci.* **307** (2003), 515–529.
- [5] E. R. Canfield, From recursions to asymptotics: on Szekeres’ formula for the number of partitions, *Electron. J. Combin.* **4**(2) (1997), #R6.
- [6] V. N. Chubarikov and G. V. Fedorov, Multiplicative and additive problems of partitions of natural numbers, in: *Continuous and Distributed Systems II*, Stud. Syst. Decis. Control **30**, Springer, Cham, 2015, pp. 29–36.
- [7] P. Erdős, On some asymptotic formulas in the theory of partitions, *Bull. Amer. Math. Soc.* **52** (1946), 185–188.
- [8] P. Erdős and J. Lehner, The distribution of the number of summands in the partitions of a positive integer, *Duke Math. J.* **8** (1941), 335–345.
- [9] F. Johansson, Arb: efficient arbitrary-precision midpoint-radius interval arithmetic, *IEEE Trans. Comput.* **66** (2017), no. 8, 1281–1292.
- [10] G. Szekeres, An asymptotic formula in the theory of partitions, *Quart. J. Math. Oxford Ser. (2)* **2** (1951), 85–108.
- [11] G. Szekeres, Some asymptotic formulae in the theory of partitions. II, *Quart. J. Math. Oxford Ser. (2)* **4** (1953), 96–111.
- [12] G. Szekeres, Asymptotic distribution of partitions by number and size of parts, in: *Number Theory, Vol. I (Budapest, 1987)*, Colloq. Math. Soc. János Bolyai **51**, North-Holland, Amsterdam, 1990, pp. 527–538.

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